

Persistent Topological Defects and Direct-Sum Time Sectors in a Relational Network Substrate:

A Formal Research Program and Toy-Model Architecture for Matter Primitives

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This paper introduces the Relational Network Substrate Framework (RNSF), a background-independent, two-layer formal research program in which smooth macroscopic spacetime and field behaviors emerge as coarse-grained descriptions of an underlying discrete relational cell complex $\mathcal{C}(\tau)$. Rather than treating matter primitives as localized point particles embedded within an independent spatial arena, RNSF models them using a persistent-homology-based descriptor, P_i , defining them as high-persistence nested cycle-complexes isolated across an embedding-free network filtration sequence $K_\bullet$.

Recent peer-reviewed advancements in persistent homology within covalent amorphous solids, knot-soliton stabilization in beyond-Standard-Model field theories, and direct-sum quantum formulations of Einstein–Rosen bridges are utilized strictly as external mathematical motivations and mechanical analogues, rather than as direct proofs of the framework’s validity. We introduce a provisional discrete linking charge ansatz, $Q_{12}^{(\Sigma)}$, evaluated over a closed, oriented three-dimensional consistency slice Σ within the evolving complex, and outline a structural junction map, $\mathcal{J}$, that maps singular Layer-1 continuous field states to orthogonal, bidirectional direct-sum Layer-2 substrate Hilbert sectors under strict norm-preserving consistency conditions.

To ensure computational verifiability, a dual-methodology toy-model algorithm is proposed, contrasting standard Vietoris–Rips simplicial filtrations against cellular cycle-basis complex pipelines to protect structural loop defects from premature topological trivialization. Finally, we explicitly detail the fundamental limitations of this v1 architecture, noting that the framework does not yet derive continuous Lorentz invariance, the Born rule, or the explicit mass and coupling spectrum of the Standard Model. RNSF is presented not as a completed theory of quantum gravity, but as a disciplined, open-source mathematical language for tracking stable topological anomalies within dynamical relational datasets.

I. INTRODUCTION

A persistent challenge across contemporary quantum gravity programs is the geometric reconciliation of continuous macroscopic spacetimes with the discrete structures hypothesized at the Planck scale [1, 2]. Standard formulations frequently encounter coordinate singularities, information-loss anomalies at causal horizons [3], and instabilities when modeling localized energy configurations without an underlying background metric [4].

The Relational Network Substrate Framework (RNSF) is a formal research program designed to address these challenges through a two-layer ontology. Layer 1 (L_1) represents the familiar, emergent domain of continuous manifolds and unidirectional field evolution. Layer 2 (L_2) represents a background-independent, purely relational ledger governed by discrete topological connectivity [5]. Rather than seeking a completed theoretical synthesis, RNSF v0.7 establishes a precise mathematical syntax to investigate whether matter-like stability and horizon information conservation can be modeled as emergent structural properties of a dynamical network.

II. RELATED WORK AND MOTIVATING PHYSICAL ANALOGUES

RNSF utilizes peer-reviewed boundary results from structural materials science, topological soliton theory, and quantum cosmology as external mathematical motivations. These works establish the plausibility of the mechanisms proposed herein.

A. Microscopic Network Topology

In smooth field descriptions, localized material responses are typically mapped to continuous density variables. However, Minamitani, Nakamura, Obayashi, and Mizuno (2025) demonstrated that the structural mechanics of non-crystalline networks are fundamentally governed by hidden, hierarchical topological layers [6]. Utilizing persistent homology on covalent amorphous solids, they found that regions of large nonaffine displacement exhibit structures where short-range disorder is embedded within medium-range ordered ring configurations, strongly correlating with localized low-energy vibrational excitations.

RNSF uses this mechanism as an analogy for defining a candidate substrate primitive. We suggest that matter-like stability may emerge when localized network irreg-

ularities are topologically trapped within medium-range structural coherence, bypassing the need for an ab initio metric description.

B. Particle-Scale Stabilization

A core problem in discrete graph models is preventing localized, loop-like defects from spontaneously dissolving under network tension. Eto, Hamada, and Nitta (2025) provided a field-theoretic analogue to this structural problem [7]. They demonstrated that stable, macroscopic knots can appear as metastable solitons within an extension of the Standard Model involving the QCD axion and right-handed neutrinos [8]. Crucially, single-species loops shrink under their own tension; stability is achieved only when distinct species are topologically linked, allowing a Chern–Simons coupling [9] to induce an internal stabilizing flux that blocks collapse.

RNSF interprets this mechanism as evidence of mechanical plausibility for defect stabilization. It mandates that an L_2 matter primitive cannot be a simple, isolated graph cycle, but must be formulated as a linked, multi-sector network defect whose collapse is energetically blocked by an internal topological charge.

C. Horizon-Scale Symmetry

At gravitational horizons or cosmological boundaries, continuous field descriptions frequently yield information-loss anomalies. Gaztañaga, Kumar, and Marto (2026) proposed a unitary reformulation of this boundary physics by introducing a direct-sum quantum framework featuring geometric superselection sectors [10]. Their model utilizes an inverted harmonic oscillator profile at phase-space horizons to describe quantum states as a direct sum of two orthogonal components possessing opposite arrows of temporal evolution, reporting significant statistical advantages when compared against large-scale Cosmic Microwave Background (CMB) parity data.

RNSF treats this framework as an external mathematical motivation for inter-layer junction physics. It suggests that when a continuous L_1 metric becomes singular or topologically ill-conditioned, information is preserved by mapping the state into the bidirectional, direct-sum relational ledger of the L_2 substrate.

III. POSTULATES OF THE FRAMEWORK

The architecture of RNSF is constructed upon five foundational postulates:

P1 The Substrate Postulate: Fundamental reality is represented by an evolving, background-independent, discrete relational cell complex

$\mathcal{C}(\tau) = (V, E, F, \dots)$, where V defines a set of discrete update events, E defines active pairwise interactions, and F defines closed relational faces. The internal sequence ordering parameter τ is non-local and distinct from macroscopic coordinate time.

P2 The Emergence Postulate: Macroscopic 4D coordinate spacetime and smooth field operators ($\mathcal{H}_{L1}[g_{\mu\nu}, \phi, A_\mu]$) are non-fundamental, coarse-grained metrics emerging asymptotically from the underlying relational connectivity and information-transfer thresholds (λ) of the complex $\mathcal{C}(\tau)$.

P3 The Topological Defect Postulate: Matter primitives are not fundamental entities residing within an independent spatial arena; they are localized, high-persistence homological defects native to the relational filtration sequence $K_\bullet$ of the substrate.

P4 The Junction Postulate: Norm preservation is imposed as a consistency condition on an isometric inter-layer junction map $\mathcal{J}$ that projects singular or horizon-bounded Layer-1 continuous states into orthogonal, bidirectional direct-sum sectors of the Layer-2 substrate Hilbert space.

P5 The Phase Transition Postulate: Global cosmological features, such as contraction-to-expansion behaviors, are modeled as occurring when the global relational update density ρ_{rel} crosses a critical structural threshold ρ_{crit} .

IV. STABILIZATION BY PROVISIONAL RELATIONAL LINKING CHARGE

To transition the concept of network knots into a formal mathematical primitive, RNSF introduces a discrete mapping of the continuum Chern–Simons linking number as a provisional ansatz.

Let $a_1 \in C^p(\mathcal{C})$ and $a_2 \in C^q(\mathcal{C})$ represent discrete relational connection cochains belonging to two independent, interacting update sectors of the global complex. We define the candidate discrete relational linking charge $Q_{12}^{(\Sigma)}$ by evaluating the cup product ($\smile$) against the coboundary operator (δ):

$$Q_{12}^{(\Sigma)} = \langle a_1 \smile \delta a_2, [\Sigma] \rangle \quad (1)$$

Where Σ is a closed, oriented 3-chain or 3-dimensional consistency slice isolated within the broader relational complex. For the present toy architecture we take $a_1, a_2 \in C^1(\mathcal{C})$, so that $\delta a_2 \in C^2(\mathcal{C})$ and $a_1 \smile \delta a_2 \in C^3(\mathcal{C})$, allowing evaluation on the closed oriented 3-chain Σ . In this formulation, $Q_{12}^{(\Sigma)}$ is introduced strictly as a candidate invariant; its conservation under localized network reconfigurations—such as discrete Pachner moves or stochastic edge-swaps—is not assumed as a completed

theorem and remains an open target condition for future computational validation.

By combining this provisional linking charge with chain-theoretic nesting rules, we formalize the structural descriptor for an RNSF particle candidate P_i :

$$P_i \equiv \left\{ [c_a] \in PH_1(K_\bullet) : \tau_a > \tau_{\text{crit}}, \exists c_b \prec c_a, Q_{12}^{(\Sigma)} \neq 0, \Delta E_{\text{unwind}} > E_{\text{barrier}} \right\} \quad (2)$$

Where $c_b \prec c_a \iff \text{supp}(c_b) \subset \text{supp}(B_a)$, given B_a is a selected 2-chain or higher-dimensional cellular support bounded by c_a ($\partial B_a = c_a$). Nesting is therefore explicitly defined as containment within the support of a bounded cellular domain, ensuring the model remains background-free. The parameter $\Delta E_{\text{unwind}} > E_{\text{barrier}}$ dictates that the relational configuration cannot dissolve into the background lattice without crossing an explicit topological energy barrier. Mass is subsequently modeled as the localized minimum relational tension E_{rel} required to maintain this frustrated configuration:

$$\text{mass} \sim \min E_{\text{rel}}(PH_k, \mathcal{N}) \quad (3)$$

V. THE JUNCTION MAPPING AND DIRECT-SUM EQUATIONS

The transition between macroscopic coordinates (L_1) and the underlying relational substrate (L_2) at a horizon boundary is governed by the junction map $\mathcal{J}$:

$$\mathcal{J} : \mathcal{H}_{L1}[g_{\mu\nu}, \phi, A_\mu] \longrightarrow \mathcal{H}_{L2}^{(+)}[K_\bullet] \oplus \mathcal{H}_{L2}^{(-)}[K_\bullet] \quad (4)$$

Norm preservation is imposed as a consistency condition on all admissible junction maps:

$$\|\Psi_{L1}\|^2 = \|\Psi_{L2}^{(+)}\|^2 + \|\Psi_{L2}^{(-)}\|^2 \quad (5)$$

This framework interprets the time-reversed, mirror quantum states identified at horizons in recent direct-sum field theories [10] not as separate physical universes, but as the activation of the bidirectional, relational ledger ($\mathcal{H}_{L2}^{(-)}$) necessary to preserve information when the macroscopic, one-way coordinate description ($\mathcal{H}_{L1}$) becomes ill-conditioned.

VI. FRAMEWORK LIMITATIONS AND UNRESOLVED DERIVATIONS

The Relational Network Substrate Framework (RNSF) is presented in this paper strictly as a formal research program and conceptual toy-model architecture. It does not constitute a complete or predictive theory of quantum gravity. The formal objects introduced—specifically the defect descriptor P_i and the linking ansatz $Q_{12}^{(\Sigma)}$ —are

mathematical proposals whose long-term stability and compatibility with physical reality remain unproven. At present, the framework does not derive, calculate, or account for:

1. **Lorentz Invariance:** The continuous spacetime symmetries of the macroscopic world have not been recovered from the non-local, discrete updates of the cell complex $\mathcal{C}(\tau)$.
2. **The Standard Model Spectrum:** The framework provides no derivation for the specific mass hierarchy, coupling constants, or three-generation flavor numbers of known leptons and quarks.
3. **The Born Rule:** The emergence of real quantum mechanical probability distributions from purely relational graph metrics remains unformulated.
4. **Path Integral Quantization:** A rigorous definition for a quantum path integral summing over distinct history configurations of an evolving relational cell complex has not been established.

VII. CONCLUSION

RNSF v0.7 introduces a persistent-homology-based descriptor for particle-like defects in a relational complex, providing an open-source mathematical language to explore topological and sector-symmetric mechanisms under a common framework. By grounding its assertions within an explicit mathematical status audit and defining comparative computational validation pipelines, the program establishes a disciplined baseline for future numerical toy-model testing.

DECLARATIONS AND AI DISCLOSURE

Portions of the manuscript organization, text formatting, and mathematical exposition were developed with AI-assisted editorial support. The underlying conceptual synthesis, verification of equations, and final authority over all scientific claims remain entirely the responsibility of the author. The manuscript was prepared in compliance with the guidelines of the ai.viXra archival repository for AI-assisted scholarly work.

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Appendix A: RNSF Mathematical Status Audit Ledger

To maintain formal accountability and protect the framework from structural overclaiming, the core algebraic objects of RNSF v0.7 are cataloged in Table I by their strict ontological status, explicit domain, and unresolved hurdles.

Appendix B: Dual-Methodology Algorithmic Toy Model Blueprint

To ensure computational verifiability, this appendix outlines a discrete, embedding-free algorithmic blueprint to simulate $\mathcal{C}(\tau)$, evaluate a relational filtration λ , and extract persistent nested topological features. To protect structural loop defects from premature topological serialization, the framework requires a comparative analysis across two distinct complex pipelines.

1. Phase 1: Graph Ledger Evolution

1. Initialize an abstract unweighted graph $G = (V, E)$, where V defines a set of discrete update events and $E \subset V \times V$ represents active primal relations.
2. Advance the internal update step τ by applying a stochastic graph-rewrite rule to simulate structural dynamics without background space.

2. Phase 2: Embedding-Free Filtration Calculation

1. For every pair of vertices (v_i, v_j) , calculate the Shortest Path Length (SPL) or Resistance Distance strictly via the topology of the ledger E .
2. Define the relational connectivity parameter λ as an integer threshold varying from 1 to $\max_diameter(G)$.
3. Construct a nested sequence of edge sets E_λ , where an edge e_{ij} is active in E_λ if and only if its calculated information-transfer cost is $\leq \lambda$. This yields the graph sequence $G_\lambda = (V, E_\lambda)$.

3. Phase 3: Comparative Complex Construction

The implementation passes the filtration data through two parallel tracks to evaluate structural preservation:

a. Track A — The Vietoris–Rips Simplicial Complex

Identify all k -cliques (fully connected subgraphs) within G_λ . Map every 3-clique to a filled 2-simplex, and

every 4-clique to a 3-simplex. This allows clean integration with standard topological libraries (e.g., ‘GUDHI’), but introduces an inherent risk of prematurely closing true structural loop defects if the localized connection density within a graph neighborhood is high.

b. Track B — The Cellular Cycle-Basis Complex

Extract a minimal fundamental cycle basis from G_λ using a cycle-finding algorithm. Construct a directed cellular complex where 2-cells are bounded strictly by these verified non-trivial cycles, applying the filtration threshold λ directly to the cycle weight vectors rather than filling higher cliques indiscriminately. This track ensures that nested, multi-generational loop features are preserved for homological tracking.

4. Phase 4: Computational Pre-Descriptor Extraction

Tracked features are extracted as un-stabilized candidates before testing phase-space constraints:

1. Compute persistent homology across the filtration sequence $K_\bullet$ to track the birth (λ_{birth}) and death (λ_{death}) of 1-dimensional loop cycles (PH_1).
2. Isolate high-persistence features where $\tau_a = \lambda_{\text{death}} - \lambda_{\text{birth}} > \tau_{\text{crit}}$.
3. Map parent-child nesting relations where $c_b \prec c_a \iff \text{supp}(c_b) \subset \text{supp}(B_a)$ and $\lambda_{\text{birth}}(c_b) > \lambda_{\text{birth}}(c_a)$.
4. Output the set of valid computational pre-descriptors:

$$P_i^{(0)} = \{[c_a] \in PH_1(K_\bullet) : \tau_a > \tau_{\text{crit}}, \exists c_b \prec c_a\} \quad (\text{B1})$$

5. Phase 5: Topological Stabilization Validation

For each extracted pre-descriptor $P_i^{(0)}$, subsequent execution blocks evaluate numerical models of $Q_{12}^{(\Sigma)}$ and boundary unwinding profiles to filter the stabilized array:

$$P_i = \left\{ P_i^{(0)} : Q_{12}^{(\Sigma)} \neq 0, \Delta E_{\text{unwind}} > E_{\text{barrier}} \right\} \quad (\text{B2})$$

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- [1] C. Rovelli, *Quantum Gravity* (Cambridge University Press, Cambridge, 2004).
 - [2] R. Loll, “Quantum gravity from causal dynamical triangulations: a review,” *Classical and Quantum Gravity*, **36**, 153001 (2019).
 - [3] S. W. Hawking, “Breakdown of predictability in gravitational collapse,” *Physical Review D*, **14**, 2460 (1976).
 - [4] T. Konopka, F. Markopoulou, and S. Severini, “Quantum graphity: a model of emergent locality,” *Physical Review D*, **77**, 104029 (2008).
 - [5] C. Rovelli, “Relational quantum mechanics,” *International Journal of Theoretical Physics*, **35**, 1637 (1996).
 - [6] E. Minamitani, M. Nakamura, K. Obayashi, and T. Mizuno, “Persistent homology elucidates hierarchical structures responsible for mechanical properties in covalent amorphous solids,” *Nature Communications*, **16**, Article number 8226 (2025). <https://doi.org/10.1038/s41467-025-63424-z>
 - [7] M. Eto, T. Hamada, and M. Nitta, “Tying Knots in Particle Physics,” *Physical Review Letters*, **135**, 091603 (2025). <https://arxiv.org/abs/2407.11731>
 - [8] R. D. Peccei and H. R. Quinn, “CP conservation in the presence of pseudoparticles,” *Physical Review Letters*, **38**, 1440 (1977).
 - [9] M. Nakahara, *Geometry, Topology and Physics* (CRC Press, Boca Raton, 2003).
 - [10] E. Gaztañaga, S. Kumar, and J. Marto, “A new understanding of Einstein–Rosen bridges,” *Classical and Quantum Gravity*, **43**, 015023 (2026). <https://doi.org/10.1088/1361-6382/ae3044>
 - [11] K. Zuse, *Calculating Space* (Massachusetts Institute of Technology, Cambridge, 1969).
 - [12] G. Carlsson, “Topology and data,” *Bulletin of the American Mathematical Society*, **46**, 255–308 (2009).
 - [13] T. Vachaspati and A. Vilenkin, “Formation and evolution of cosmic strings,” *Physical Review D*, **30**, 2036 (1984).
 - [14] E. Witten, “Quantum field theory and the Jones polynomial,” *Communications in Mathematical Physics*, **121**, 351–399 (1989).
 - [15] S. Weinberg, “The cosmological constant problem,” *Reviews of Modern Physics*, **61**, 1–23 (1989).

TABLE I. RNSF v0.7 Mathematical Status Audit Ledger

Object	Operational Domain	Scholarly Status	Unresolved Hurdle / Future Target
$\mathcal{C}(\tau)$	Discrete cell complex	Postulate	Deriving a continuous 4D spacetime manifold asymptotic limit.
λ / K_λ	Scalar field mapping to $\mathcal{C}$	Technical Definition	Systematically proving stability under random node variations.
$PH_k(K_\bullet)$	Topological data spaces	Rigorous Tool	Quantifying the exact noise floor for computational graph variations.
$Q_{12}^{(\Sigma)}$	Cochain evaluation mapping	Provisional Ansatz	Explicitly deriving the local update moves that preserve this invariance.
P_i	Filtered cycles in $PH_1(K_\bullet)$	Defect Descriptor	Mapping these persistence spectrum states to known particle numbers (spin, ...)
$\mathcal{J}$	Inter-space mapping	Proposed Mapping	Formally deriving the exact boundary metrics that preserve the norm equation.